

PROJECT REPORT OF CO-OP

Project at Industry (Module-2)

ON

Unlocking the Potential of India's Insurance Industry: Growth Drivers, Challenges, and Insurtech Pathways

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Submitted by:

Shaurya Mor (2210992295)

Sehdev Singh Saini (2210992279)

Eva Aggarwal (2210991556)

Supervised By:

Dr. Ashish Kumar

Associate Professor



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

CONTENTS

S.No.	Title	Page No.
1	Declaration	ii
2	Acknowledgement	iii
3	Abstract	1
4	Introduction	1
5	Methodology	3
6	Result and Discussions	5
7	Conclusion	6
8	References	9
9	Appendices	10

DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Unlocking the Potential of India's Insurance Industry: Growth Drivers, Challenges, and Insurtech Pathways" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Ashish Kumar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

(Shaurya Mor, Sehdev Singh Saini, Eva Aggarwal)

Date: April 29, 2026

Roll No. 2210992295, 2210992279, 2210991556

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Ashish Kumar

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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Abstract

The insurance industry represents an important element of financial stability and risk management. In India, although being considered one of the biggest emerging countries with a large population, the degree of insurance penetration is rather low compared to international standards. At the moment, it is estimated that insurance penetration in India is about 3.7% of the country's GDP, which is noticeably lower than the world average, i.e., 7%. In addition, there are some concerns related to insurance density, which is about USD 92 per capita compared to USD 900 across the world. Therefore, this paper discusses the Indian insurance industry by analyzing its current state of affairs, key structural problems faced by this industry, and main factors of growth. Alongside with this, the study addresses the concept of InsurTech and its impact on insurance innovation.

Keywords: Insurance, InsurTech, IRDAI, India, Health Insurance, Life Insurance, Embedded Insurance, Digital Infrastructure, Insurance Penetration

1. Introduction

Insurance can be regarded as a financial tool created to serve as a form of protection against risks through which people pay premiums for receiving coverage in case of unexpected happenings. This is one of the oldest financial instruments that are widely accepted and used in the modern world in the course of financial planning. The insurance industry in India, which is home to almost one-sixth of the planet's population, has a lot of room for development despite some positive indicators.

The last two decades have seen many changes in the Indian insurance market. Until 1999, the market was mainly monopolized by the state, the Life Insurance Corporation of India being the only player in life insurance business, while other sectors were dominated by several government-owned companies. The liberalization process initiated in 1999 and the creation of IRDAI as an independent regulatory body marked a new phase of the market's development. As a result, today there are many more players in the market, more insurance products available, and new types of distribution channels.

Nowadays, there is a tendency towards technology-oriented firms entering the insurance market. By using digital technologies like smartphones, artificial intelligence, and the country's payment infrastructure, these companies seek to expand their business, improve service quality, increase efficiency, and make insurance products available and affordable for everyone. Companies like Plum Insurance are even helping small and medium-sized businesses provide adequate health

insurance for their workers.

This paper considers the Indian insurance industry through three

major perspectives, namely, the current situation on the market, factors hindering its further development, and reasons behind growth rates. Further discussion of this topic will include a review of the market's dynamics, barriers to buying insurance products, growth factors, and the contribution made by InsurTech firms to its development.

2.1 Insurance Penetration and Market Development in India

Penetration of insurance is a major issue in India. The Insurance Regulatory and Development Authority of India (IRDAI) noted that there was an insurance penetration rate of around 3.7% of GDP in 2024, which is well below the world average of around 7%. The larger portion of the total premium in India relates to life insurance.

Swiss Re Institute predicts that India's insurance premiums will rise at an annual growth rate of 7.1% between 2023 and 2028, making it one of the fastest-growing insurance markets amongst G20 countries. However, insurance density is still low, at around USD 92 per capita per year, compared to highly developed nations like the UK.

According to the Indian Brand Equity Foundation (IBEF), the growing income levels of the middle class and increased awareness are likely to accelerate market growth. Likewise, the IMARC Group expects robust market growth up until 2034, owing to liberalization and heightened awareness amongst consumers.

2.2 Structural Barriers to Insurance Uptake

Several studies have found various structural factors that inhibit insurance take-up in India. The lack of parity between rural and urban areas is one of them. According to the data from the National Sample Survey Office, only around 22% of rural Indian households have insurance coverage. The distribution system, which includes agents, branches, and online sites, is heavily skewed towards urban areas.

The language issue in policy documents is another crucial determinant. Insurance policies are generally written in English or Hindi, thus creating barriers in a linguistically heterogeneous country like India.

Finally, trust issues and negative claims experiences are also examples of demand-side barriers. While the claim settlement ratio is reported to be relatively high based on the information provided by the IRDAI, it should be noted that only those claims which are filed are considered.

2.3 Growth Drivers: Demographics, COVID-19, and Digital Infrastructure

A number of factors have been found to be driving insurance penetration from the demand side. The youth in India is expected to be a driver in the coming years because younger consumers are more likely to purchase insurance policies. According to Boston Consulting Group (BCG) and the Federation of Indian Chambers of Commerce and Industry (FICCI), India's middle-class population is expected to reach 600 million by 2035.

The health sector experienced an increase in insurance demand due to the coronavirus outbreak. The IRDAI data shows that health insurance premiums increased by around 35% during FY2021. Furthermore, according to Grand View Research, the health insurance market is expected to continue growing from 2025 to 2030.

Technology advancements in India have been another factor facilitating insurance development. IBEF stated that the adoption of technology systems like Aadhaar, Unified Payments Interface (UPI), and Ayushman Bharat Digital Mission reduced barriers to entry and streamlined claims processing.

2.4 Regulatory Reform and the IRDAI Vision 2047 Framework

The literature discusses a phase of regulatory reform through IRDAI's guidance. Through its guidance, IRDAI has enabled higher foreign involvement in the industry by permitting up to 100% FDI in the insurance sector.

Under the 'Use and File' regulation, insurers can introduce retail insurance products without seeking prior regulatory approvals, which enhances product innovation.

IRDAI Vision 2047 aims to achieve universal coverage of the insurance sector in India. BCG and FICCI reports suggest that InsurTech-driven microinsurance and embedded insurance models will play a vital role in this process.

2.5 InsurTech: Distribution, Automation, and Embedded Models

There has been an increased number of academic studies on InsurTech developments in India after 2020. According to the findings of BCG and FICCI, there were more than 150 InsurTech companies in India, valued at around \$13.6 billion.

Digital aggregators such as Policybazaar have made the process of accessing insurance data by customers easier and raised competition among insurance companies. In addition, digital-only insurers such as Plum Insurance have provided solutions for small and medium-sized businesses

for lack of coverage in health insurance.

Other emerging areas of academic research are artificial intelligence in claims management through image-based claims settlements in motor claims and AI-enabled customer service centers in multilingual environments.

2.6 Research Gap

The current literature offers valuable perspectives on macro-level market dynamics and new technology-based developments in InsurTech. Yet, it fails to offer an all-encompassing understanding of the interconnection between the factors of distrust, innovation in distribution, and regulatory support. This paper attempts to fill the void by examining the interplay among these variables in the Indian insurance market.

2. Methodology

2.1 Research Approach

This research looks at the Indian insurance industry with an aim of understanding the present state of market structure, ascertaining some of the factors that affect its growth and evaluating the impact of InsurTech in reshaping the insurance industry. It follows a descriptive and analytical methodology and does not produce predictions.

This research uses secondary data only. Data was gathered from various published documents produced by official government agencies and industry associations. The methodology did not involve any kind of primary data gathering through surveys and interviews.

2.2 Data Sources and Coverage

The data used in the research is obtained from eight major sources. They include regulatory reports, government surveys, and industry research reports. The sources give an overview of the market's performance, structure, and recent trends.

These sources of data include regulatory reports, government surveys, and industry research reports. Table 1 and Table 2 list the different sources of data, the scope of each source, and the period covered.

2.3 Study Period

The study mainly focuses on data collected between FY2018-FY2023. This time frame was chosen to ensure the analysis covered three different stages: the pre-COVID stage, the COVID stage, and the post-COVID recovery stage.

Where necessary, more updated data is included to offer an up-to-date view of trends in the industry.

3. RESULTS AND DISCUSSION

The results in the current section provide a summary of the most important results of the research in four areas: general market growth and benchmarking; structural obstacles in obtaining insurance; the effect of the COVID-19 crisis on health insurance; and the growth of the InsurTech ecosystem. These results will be provided in tables and charts based on the information sources described in the Methodology section.

3.1 Market Growth and Global Benchmarking

There has been consistent growth within India's insurance industry over the years. Insurance premium income increased to ₹7.05 lakh crore (USD 82 billion) in FY2024-25, showing an overall growth of 5.6% over the previous fiscal year. Life insurance made up approximately 75% of the premiums generated.

Compound Annual Growth Rate (CAGR) for the period between FY2018-FY2023 stood at 11% for life insurance premiums, which is considerably higher compared to the global average CAGR of 4%.

Nonetheless, even though India has been experiencing substantial growth, its insurance coverage, both penetration (3.7% of GDP) and density (USD 92 per capita), is much lower than global standards.

3.2 Claim Settlement Ratios and the Trust Barrier

The claim settlement ratio is high in case of major players in life insurance, varying between approximately 96.69% to 99.65%, with LIC settling 98.76%. As indicated in table 4 (Page 3), these ratios have continued to improve over the years.

It is worth noting, however, that the above statistics refer to settled claims and ignore a wider audience of customers who are unwilling to file a claim owing to expected denial of such claims

in light of previous experience. There is a tendency of overestimation of consumer experience. The results indicate that lack of trust is a major impediment to insurance rather than cost, as illustrated by statistics that about 22% of rural citizens purchase insurance, despite rural citizenry forming the majority.

3.3 COVID-19 and the Structural Shift in Health Insurance Demand

The most critical change on the demand side is health insurance. It is noteworthy that the volume of premiums continued increasing after the end of the pandemic, and therefore, it can be said that this trend demonstrates a fundamental change in consumers' preferences, not just an anomaly. Consequently, such a phenomenon had a considerable impact on distribution channels since a substantial number of new customers acquired health insurance via digital means.

3.4 InsurTech Growth and Ecosystem Development

In terms of growth, India's InsurTech landscape has undergone significant changes over the period studied. As per Table 5 (page 4) and Figure 4 (page 5), the sector has seen an increase in its revenues by approximately 12 times over the last five years, positioning itself as the second-biggest InsurTech marketplace in Asia Pacific.

Technology-enabled approaches have contributed to the tangible improvements in the following spheres:

- Use of digital aggregators like Policybazaar helped reduce information asymmetries due to standardized product comparisons.
- Digital insurers such as Plum Insurance made group health insurance policies more easily available to SMEs.
- Use of AI in claims automation has shortened the settlement period up to several hours from days or even weeks.

Additionally, India's well-developed digital environment has made InsurTech solutions even more effective as far as identity verification and payments are concerned.

3.5 Discussion

Combined, the results lead to several major findings.

First, even though India's insurance market is developing at a faster pace than the world average in absolute terms, the continuing difference in penetration and density suggests that the structural barriers, not financial, play the crucial role.

Second, trust, in particular, trust impacted by the claim experience proves to be a greater challenge than affordability for many individuals.

Third, there is an unambiguously increasing demand for health insurance due to the COVID-19 crisis and increased utilization of online distribution systems.

Finally, InsurTech, which can be viewed as a new way of distributing policies, provides a comprehensive approach aimed at solving inefficiency problems of the conventional insurance, such as the asymmetry of information, low accessibility, and inefficient claims handling.

There is one important conclusion about the unmet demand that should be made as well; a considerable part of the rural population remains without insurance. It is necessary to come up with some additional measures regarding product customization, last-mile delivery, and other issues related to inconsistent incomes.

4. CONCLUSION

The Indian insurance industry has plenty of room for further development, which is facilitated by the large number of people, widespread use of smartphones, and fast expansion of the digital payment system. The introduction of all those elements has greatly contributed to the availability of insurance services and made processes related to collecting premiums and processing claims much easier.

Nonetheless, according to the results obtained during the course of this research, one can conclude that problems with low coverage of insurance services in the region have little to do with cost barriers. In contrast, the reasons for such outcomes include such factors as lack of consumer trust, low levels of product awareness, and inefficiencies in the process of distribution. InsurTechs in India have been making great strides in improving those aspects through simplifying products, providing support via regional languages, implementing innovations in digital distribution of insurance products, and optimizing the claims processing process. Not only do those developments contribute to the commercial success of InsurTech, but they also have profound socio-economic value due to possible negative consequences of a lack of insurance. Thus, the further development of the insurance industry in India requires the joint efforts of various stakeholders, including insurance companies, technology providers, and government bodies. With all those measures being implemented properly, India can become a highly developed insurance market in the future.

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Appendices

Appendix A — International Benchmarking Data (India vs. US, UK, China, South Africa, etc. on penetration and density)

Appendix B — Regulatory History and Policy Timeline (key IRDAI milestones from 1999 to the 2047 "Insurance for All" framework)

Appendix C — Insurtech Ecosystem Profiles (Policybazaar, Acko, Digit, Plum, Coverfox with valuations and innovations)

Appendix D — Distribution Channel Analysis (agents, bancassurance, digital, brokers, micro-insurance with market share estimates)

Appendix E — Research Methodology Notes (study design, data sources, temporal scope, and limitations)

Appendix F — Glossary of Key Terms (14 definitions covering IRDAI, UPI, Bima Sugam, CAGR, BOP, and more)